信号与系统实验

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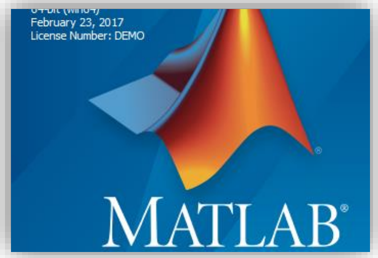

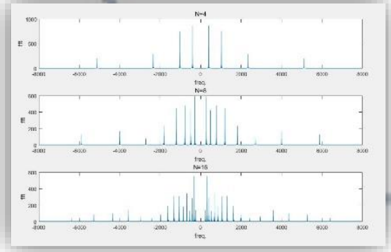

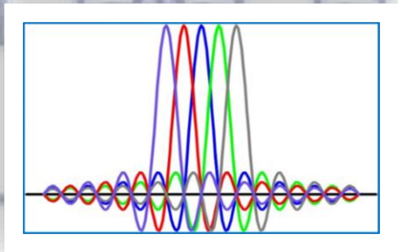

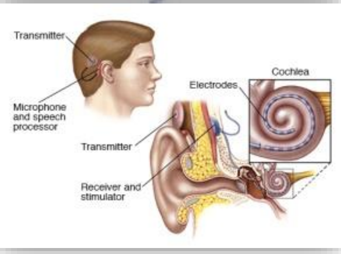

Southern University of Science and Technology

Signals and Systems (Lab)

Lab 1: MATLAB Programming

Part 3

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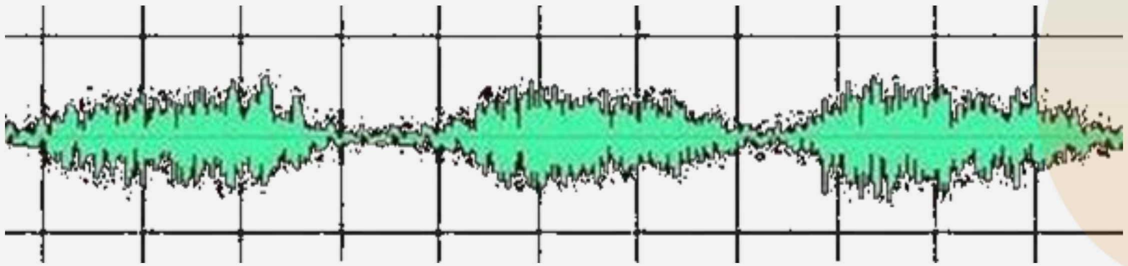
1. Objective：Analysis in frequency-domain

The music signal can be represented by two different ways.



Time-domain

Frequency-domain

Citizen Center, Shenzhen

5.1 Preliminary knowledge

Variable — Every variable is a matrix

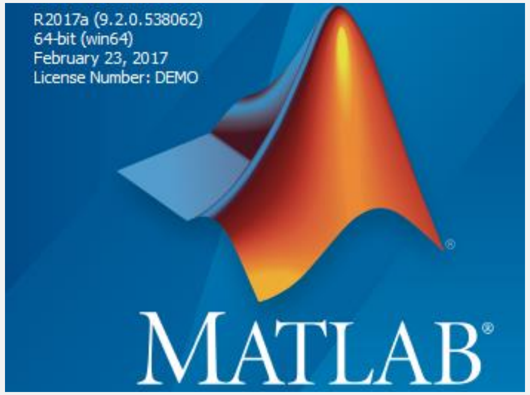
MATLAB Program.

Operation — Matrix-based operation

Sub function

Flow control — If, for, while, break, …

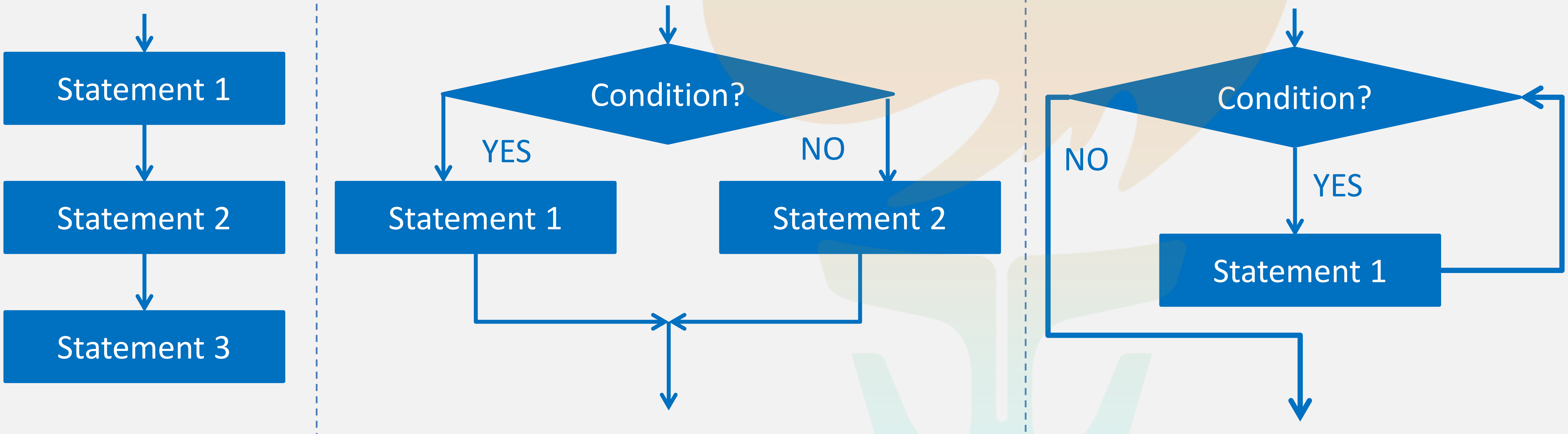
Linear algebra



7.2 Flow Control

MATLAB program has three basic structures:

- Sequence, Selection and Repetition



Sequence **Selection: if** **Repetition: for & while**

7.9 Lab Assignments

■ 1.4 Properties of Discrete-Time Systems

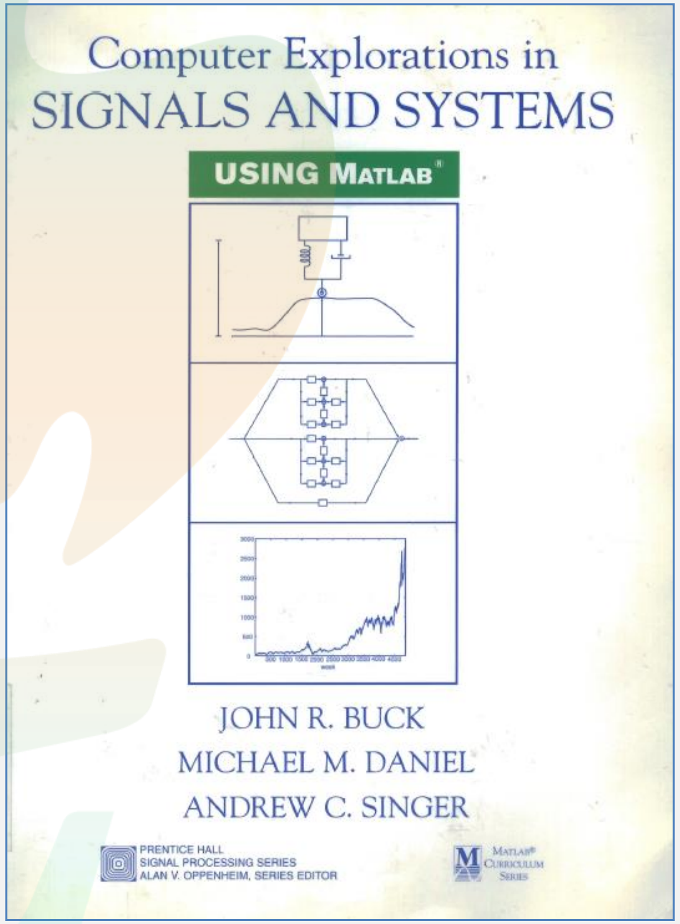
Discrete-time systems are often characterized in terms of a number of properties such as linearity, time invariance, stability, causality, and invertibility. It is important to understand how to demonstrate when a system does or does not satisfy a given property. MATLAB can be used to construct counter-examples demonstrating that certain properties are not satisfied. In this exercise, you will obtain practice using MATLAB to construct such counter-examples fcr a variety of systems and properties.

Basic Problems

For these problems, you are told which property a given system does not satisfy, and the input sequence or sequences that demonstrate clearly how the system violates the property. For each system, define MATLAB vectors representing the input(s) and output(s). Then, make plots of these signals, and construct a well reasoned argument explaining how these figures demonstrate that the system fails to satisfy the property in question.

(a). The system $y[n] = \sin((\pi/2)x[n])$ is not linear. Use the signals $x_1[n] = \delta[n]$ and $x_2[n] = 2\delta[n]$ to demonstrate how the system violates linearity.

(b). The system $y[n] = x[n] + x[n+1]$ is not causal. Use the signal $x[n] = u[n]$ to demonstrate this. Define the MATLAB vectors **x** and **y** to represent the input on the interval $-5 \leq n \leq 9$, and the output on the interval $-6 \leq n \leq 9$, respectively.



8.1 Basic task: Plot the signal sin(x) between 0≤x≤4π

PlotCommand.mlx

Create an x-array of 100 samples between 0 and 4π

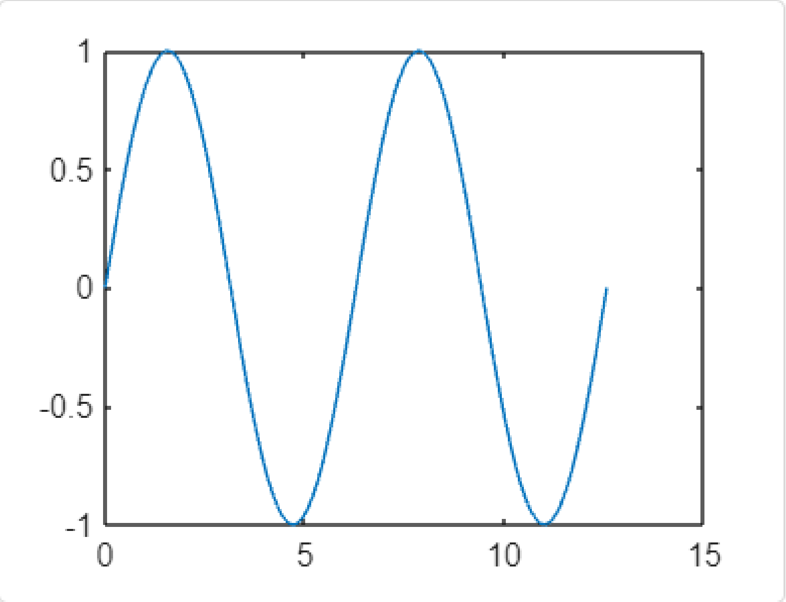
```
x=linspace(0,4*pi,100);
```

Calculate sin(.) of the x-array

```
y=sin(x);
```

Plot the y-array

```
plot(x,y)
```



8.1 Plot the signal e-x/3sin(x) between 0≤x≤4π

PlotCommand2.mlx

Create an x-array of 100 samples between 0 and 4π.

```
x=linspace(0,4*pi,100);
```

Calculate sin(.) of the x-array

```
y=sin(x);
```

Calculate e-x/3 of the x-array

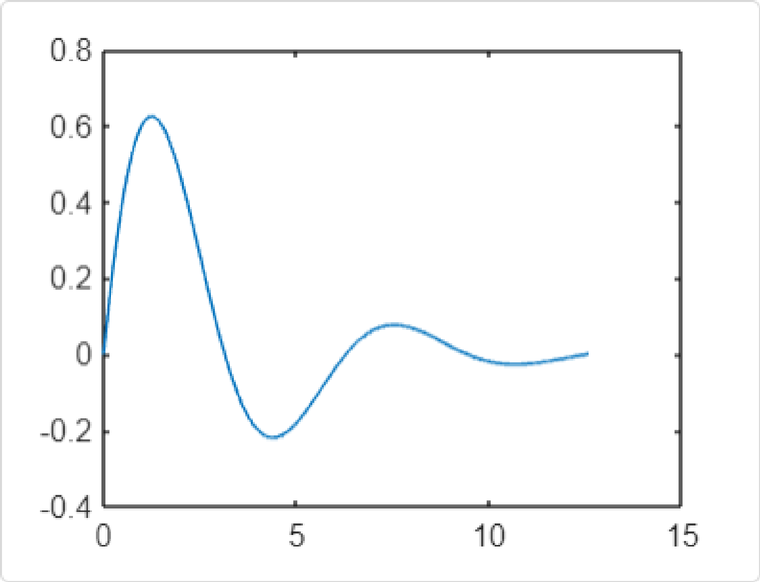
```
y1=exp(-x/3);
```

Multiply the arrays y and y1

```
y2=y.*y1;
```

Plot the y2-array

```
plot(x,y2)
```



8.2 Plot and Stem

PlotCommand.mlx *

Create an x-array of 100 samples between 0 and 4π

```
x=linspace(0,4*pi,100);
```

Calculate sin(.) of the x-array

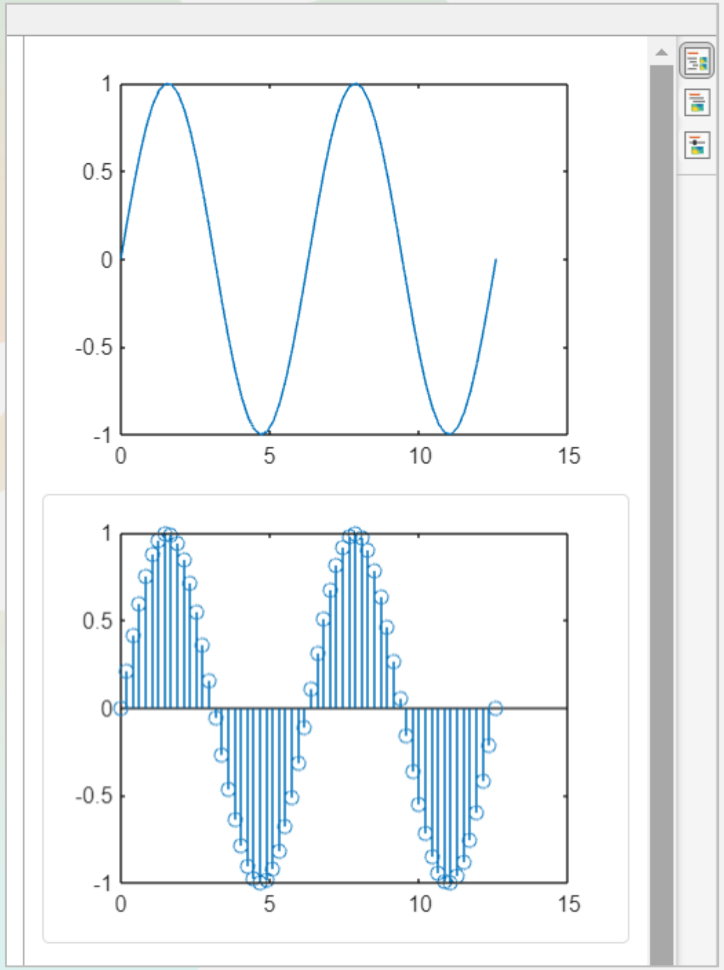
```
y=sin(x);
```

Plot the y-array

```
plot(x,y)
```

Stem the y-array

```
stem(x,y)
```



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8.3 title,xlabel and ylabel

PlotCommand.mlx

Create an x-array of 100 samples between 0 and 4π

```
x=linspace(0,4*pi,100);
```

Calculate sin(.) of the x-array

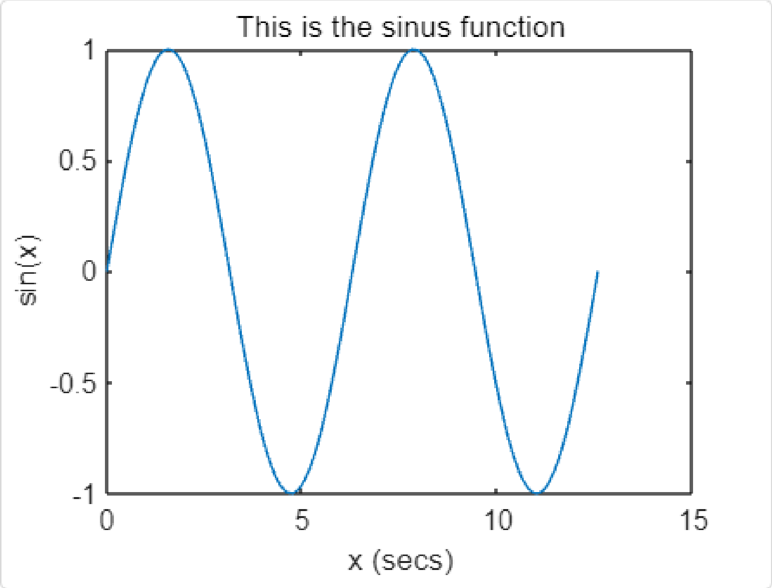
```
y=sin(x);
```

Plot the y-array

```
plot(x,y)
```

Title

```
title('This is the sinus function')
xlabel('x (secs)')
ylabel('sin(x)')
```



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8.4 Line types, plot symbols and colors

- Plot with various line types, plot symbols and colors

Colors		Symbols		Line types	
b	blue	.	point	-	solid
g	green	o	circle	:	dotted
r	red	x	x-mark	-.	dashdot
c	cyan	+	plus	--	dashed
m	magenta	*	star	(none)	no line
y	yellow	s	square		
k	black	d	diamond		
w	white	v	triangle (down)		
		^	triangle (up)		
		<	triangle (left)		
		>	triangle (right)		
		p	pentagram		
		h	hexagram		

8.4 Example

PlotCommand3.mlx

Create an x-array of 100 samples between 0 and 4π

```
x=linspace(0,4*pi,100);
```

Calculate sin(.) of the x-array

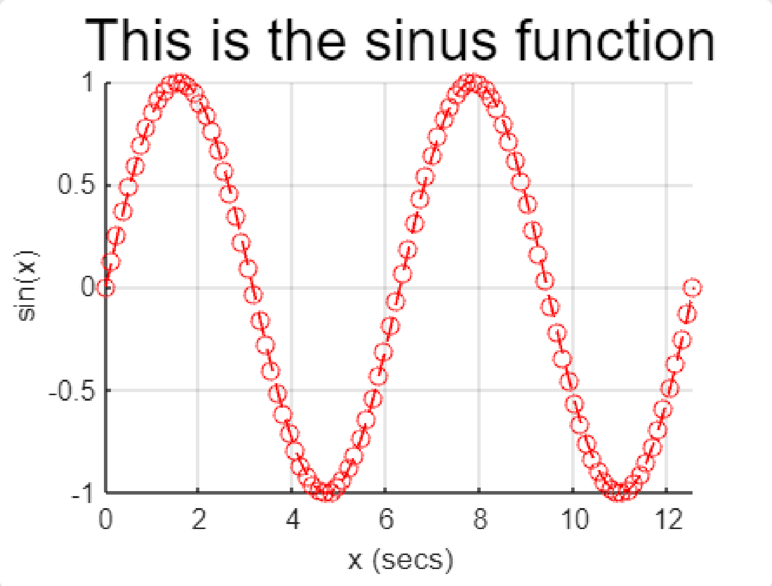
```
y=sin(x);
```

Plot the y-array

```
plot(x, y, 'r--o')
grid on
box off
axis([0 4*pi ylim])
```

Title

```
title('This is the sinus function','fontsize',20)
xlabel('x (secs)')
ylabel('sin(x)')
```



8.5 More plots in one panel

PlotCommand4.mlx

Create an x-array of 100 samples between 0 and 4π

```
x=linspace(0,4*pi,100);
```

Calculate sin(.) of the x-array

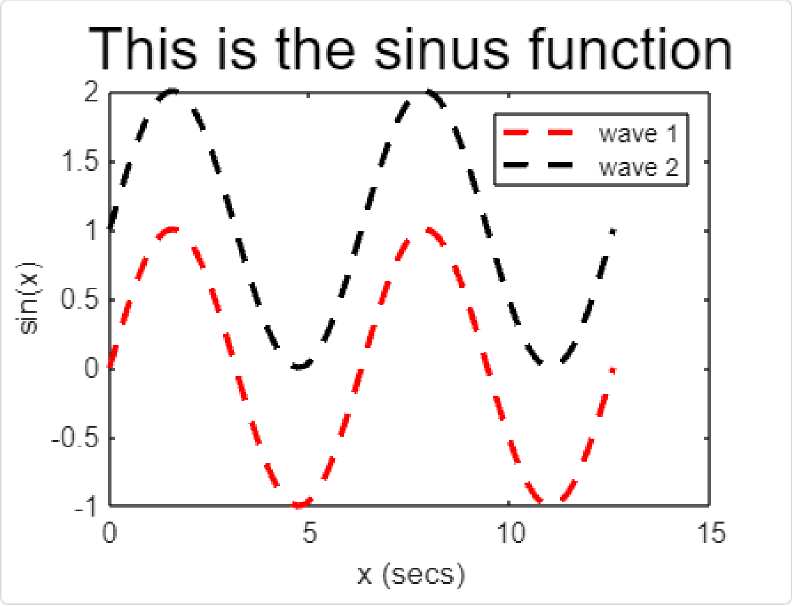
```
y=sin(x);
```

Plot the y-array

```
figure(1)
plot(x, y, 'r--', 'linewidth',2)
hold on
plot(x, y+1, 'k--', 'linewidth',2)
legend('wave 1', 'wave 2')
```

Title

```
title('This is the sinus function','fontsize',20)
xlabel('x (secs)')
ylabel('sin(x)')
```



8.6 subplot

PlotCommand5.mlx *

Create an x-array of 100 samples between 0 and 4π

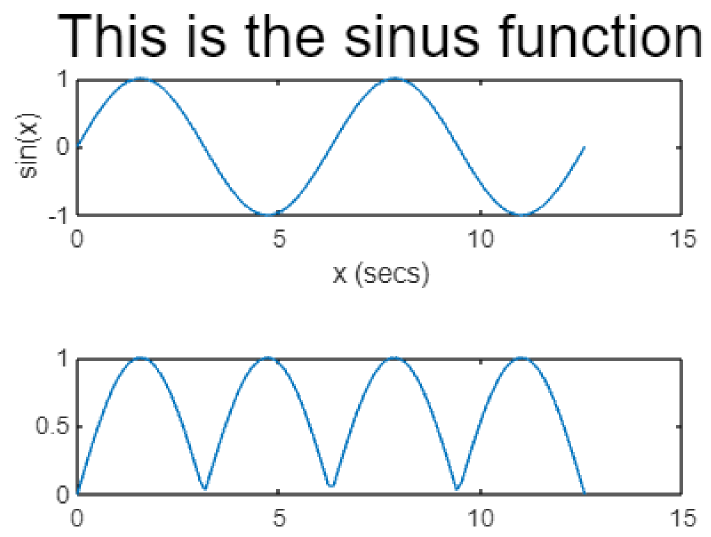
```
x=linspace(0,4*pi,100);
```

Calculate sin(.) of the x-array

```
y=sin(x);
```

Plot the y-array

```
figure(1)
subplot(2,1,1),plot(x,y)

title('This is the sinus function','fontsize',20)
xlabel('x (secs)')
ylabel('sin(x)')

subplot(2,1,2),plot(x,abs(y))
```



8.7 Some useful functions

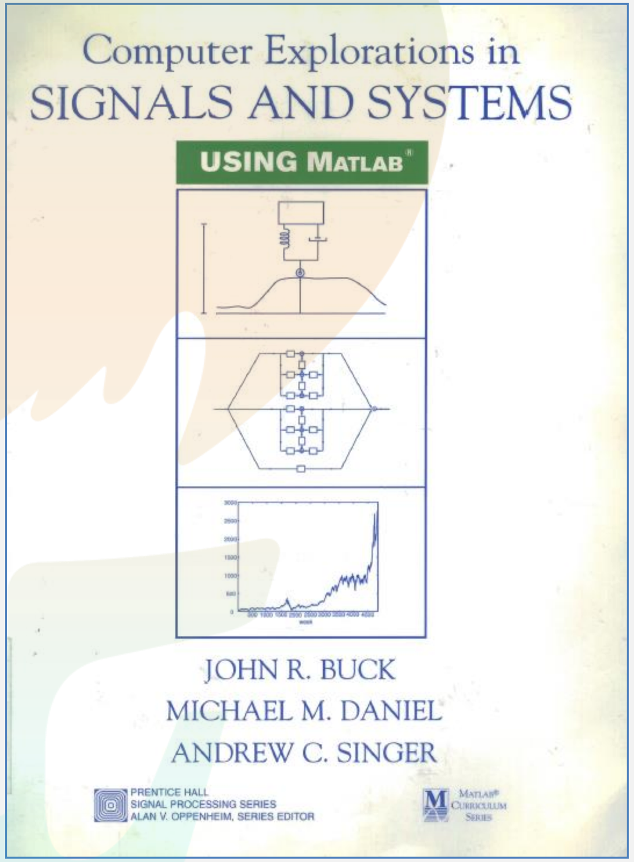
- length(.)
- size(.)
- abs(.)
- sum(.)
- mean(.)
- std(.)
- diff(.)
- ……

命令行窗口

```
>> help mean
 mean - 数组的均值
    此 MATLAB 函数 返回 A 沿大小不等于 1 的第一个数组维度的元素的均值。

    M = mean(A)
    M = mean(A,'all')
    M = mean(A,dim)
    M = mean(A,vecdim)
    M = mean(___,outtype)
    M = mean(___,nanflag)

    输入参数
        A - 输入数组
            向量 | 矩阵 | 多维数组
        dim - 沿其运算的维度
            正整数标量
        vecdim - 维度向量
            正整数向量
        outtype - 输出数据类型
            'default' (默认值) | 'double' | 'native'
        nanflag - NaN 条件
            'includenan' (默认值) | 'omitnan'

fx  打开示例
```

8.8 Lab Assignments

- 1.4(a)(b)(c)(d)
- 1.5(a)(b)(c)
- Submit your report.



Question ?